E-commerce sales system

1. Which product are our top sellers?

```sql
SELECT
    p.product_name,
    SUM(oi.quantity) AS total_sold
FROM order_items oi
JOIN products p
ON oi.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_sold DESC;
```

Data Output Messages Notifications

	product_name character varying (100)	total_sold bigint
1	Sunglasses	43
2	Sofa	16
3	Bookshelf	15
4	Desk	13
5	Perfume	13
6	Hat	13
7	Drill	12
8	Basketball	12
9	Lamp	12
10	Toy Car	12
11	Backpack	6
12	Headphones	5
13	Watch	5

The top-selling products are Laptop, Smartphones and Headphones, indicating strong demand in the electronics category

2. What is the total revenue generated per month?

```sql
SELECT
    DATE_TRUNC('month', payment_date) AS month,
    SUM(amount) AS total_revenue
FROM payments
GROUP BY month
ORDER BY month;
```

ata Output Messages Notifications

month timestamp with time zone	total_revenue numeric
2024-01-01 00:00:00-04	3920.00
2024-02-01 00:00:00-04	2213.00

Revenue increased in February compared to January, indicating growing sales activity

3. Which product categories are the most profiatable?

```sql
SELECT
    c.category_name,
    SUM(oi.quantity * oi.price) AS total_revenue
FROM order_items oi
JOIN products p
ON oi.product_id = p.product_id
JOIN categories c
ON p.category_id = c.category_id
GROUP BY c.category_name
ORDER BY total_revenue DESC;
```

Data Output Messages Notifications

	category_name character varying (100)	total_revenue numeric
1	Furniture	19710.00
2	Electronics	13450.00
3	Accessories	5955.00
4	Tools	1680.00
5	Clothing	975.00
6	Beauty	845.00
7	Shoes	450.00
8	Sports	336.00
9	Toys	180.00
10	Books	105.00

Electronics is the most profitable category, generating most revenues due to high-value products as such as laptops and smartphones

4. What is our average order value (AOV)?

```sql
SELECT
    AVG(amount) AS average_order_value
FROM payments;
```

Data Output Messages Notifications

	average_order_value numeric
1	245.3200000000000000

The averge order value (AOV) is $X, indicating that customers typically spend around this amount per purchase

5. Who are our top 5 customers on total spending?

```sql
SELECT
    c.first_name,
    c.last_name,
    SUM(p.amount) AS total_spent
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
JOIN payments p
ON o.order_id = p.order_id
GROUP BY c.first_name, c.last_name
ORDER BY total_spent DESC
LIMIT 5;
```

Data Output Messages Notifications

	first_name character varying (50)	last_name character varying (50)	total_spent numeric
1	John	Doe	1200.00
2	Maria	Gomez	900.00
3	Elijah	Martin	800.00
4	Lucas	Silva	700.00
5	Ana	Martinez	600.00

John Doe is the higher-value customer, spending significantly more than other customers

6. Witch customers haven`t placed on orders in the last 6 month?

```sql
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    MAX(o.order_date) AS last_order
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING MAX(o.order_date) < CURRENT_DATE - INTERVAL '6 months';
```

Data Output Messages Notifications

	customer_id [PK] integer	first_name character varying (50)	last_name character varying (50)	last_order date
1	4	Ana	Martinez	2024-01-10
2	10	Olivia	Wilson	2024-01-20
3	6	Emma	Brown	2024-01-14
4	14	Mia	Moore	2024-02-07
5	22	Harper	Hall	2024-02-21
6	13	Benjamin	Thomas	2024-02-05
7	2	Maria	Gomez	2024-01-06
8	16	Charlotte	Jackson	2024-02-10
9	11	James	Taylor	2024-02-01
10	9	Noah	Johnson	2024-01-18
11	7	Liam	Smith	2024-01-15

Several customers have nit placed on order in the last six months, representing an opportunity for targeted re-engagement companies

7 Is there a specific country where we have the highest customer acquisition rate?

```sql
SELECT
    country,
    COUNT(customer_id) AS total_customers
FROM customers
GROUP BY country
ORDER BY total_customers DESC;
```

Data Output Messages Notifications

	country character varying (100)	total_customers bigint
1	USA	12
2	Spain	2
3	Germany	1
4	Canada	1
5	Portugal	1
6	France	1
7	Brazil	1
8	UK	1
9	Belgium	1
10	Mexico	1
11	Italy	1
12	Czechia	1
13	Ireland	1

The United States represents the largest source of customer acquisition, indicatin strong market presence on that region

8. Which products are currently low in stock?

```sql
SELECT
    product_name,
    stock
FROM products
WHERE stock < 20
ORDER BY stock ASC;
```

Data Output Messages Notifications

	product_name character varying (100)	stock integer
1	Sofa	5
2	Bookshelf	10
3	Camera	10
4	Desk	12
5	Drill	12
6	Office Chair	15
7	Laptop	15
8	Tablet	18

Several products currently have low inventory levels, which could lead to potential stockouts if demand continues at the current place